

Skill - 16

1. Background Jobs and Job Table

```
harshika@siri: ~$ gedit skill16.c

** (gedit:7601): WARNING **: 03:43:59.486: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:7601): WARNING **: 03:43:59.487: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill16.c -o skill16
harshika@siri:~$ ./skill16
Background Job Shell - Skill 16
Use '&' for background jobs.
Use 'jobs' to display job table.
Use 'cleanup' to remove completed jobs.
Type 'exit' to quit.

myshell> sleep 10 &
[1] 8030 Running sleep 10
myshell> jobs

Job Table:
ID    PID    STATUS    COMMAND
1      8030    Running    sleep 10

myshell> sleep 5 &
[2] 8037 Running sleep 5
myshell> jobs

Job Table:
ID    PID    STATUS    COMMAND
1      8030    Completed  sleep 10
2      8037    Running    sleep 5

myshell> exit
Shell exited.
harshika@siri:~$ gedit skill16.c

** (gedit:8049): WARNING **: 03:45:03.838: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:8049): WARNING **: 03:45:03.838: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/wait.h>
6 #include <signal.h>
7
8 #define INPUT_SIZE 300
9 #define MAX_JOBS 20
10
11 struct Job
12 {
13     int job_id;
14     pid_t pid;
15     char command[200];
16     char status[20];
17 };
18
19 struct Job jobs[MAX_JOBS];
20 int job_count = 0;
21 int next_job_id = 1;
22
23 /* Add a new background job */
24 void add_job(pid_t pid, char *command)
25 {
26     if (job_count >= MAX_JOBS)
27     {
28         printf("Job table is full.\n");
29         return;
30     }
31
32     jobs[job_count].job_id = next_job_id++;
33     jobs[job_count].pid = pid;
34
35     strcpy(jobs[job_count].command, command);
36     strcpy(jobs[job_count].status, "Running");
37
38     printf("[%d] %d Running %s\n",
39           jobs[job_count].job_id,
40           jobs[job_count].pid,
41           jobs[job_count].command);
42
43     job_count++;
44 }
45
46 /* Check background jobs */
47 void update_jobs()
48 {
49     for (int i = 0; i < job_count; i++)
50     {
51         if (strcmp(jobs[i].status, "Running") == 0)
52             continue;
53     }
54 }
```